Collaborative Filtering Report

1. Data Preparation

- We used a preprocessed reviews dataset containing user ratings (user_id, book_id, ratings).
- Constructed a user-item interaction matrix where each row represents a user, each column a book, and the values are ratings.
- We constructed a user-item interaction matrix using the pivot_table() method,
- We built a matrix where:
  - Rows = users
  - Columns = books
  - Values = user ratings
- Missing values were filled with 0, indicating no interaction.
- Shape: (8423 users × 2206 books)
  Sparsity: Very high (most users rate only a few books)

2. User-User Collaborative Filtering

- Used the cosine similarity metric with KNN (k=6) to find similar users.
- For a target user, we retrieved books liked by similar users but not yet rated by the user.

Performance:

- Worked only for users with multiple overlapping ratings.
- Most users received no recommendations due to sparsity.

Example Output:

```
User-User Recommendations for User A3HHXRELK8BHQG: ['B002HJV4PM', 'B002YX0PL0']
User-User Recommendations for User A2RGNZ0TRF578I: []
User-User Recommendations for User A3S0H2HV6U1I7F: []
User-User Recommendations for User AC4OQW3GZ919J: []
User-User Recommendations for User A3C9V987IQHOQD: []
```

3. Item-Item Collaborative Filtering

- Compares books by co-rating similarity.
- If you liked Book A, we recommend books rated similarly by other users.

Strength:

- Performs well even if user rated only 1 book.
- Better than User-User CF for sparse data.
- Could recommend books for users who rated even one book if that book had strong similarity scores with others.

Example Output:

```
User A H Kobayashi => Item-Item Recommendations: []
User A K P => Item-Item Recommendations: ['553508784', '110199715X', '545931908', '1250787653', '1635575583']
User A Reviewer => Item-Item Recommendations: ['553508784', '545931908', '1250787653', '1450805752', '1635575583']
User A Slater => Item-Item Recommendations: ['006332752X', 'B003VYBP9M', 'B003VTZTUI', 'B003VWCCTQ', 'B003VYAY6C']
User A0089401235VSN3Z6F3HK => Item-Item Recommendations: ['B0036Z9YEE', 'B00427ZJ1C', 'B003OIBGSU', 'B002RKRMSY', 'B002RKSZJO']
```

4. Matrix Factorization (SVD)

- Applied TruncatedSVD (20 components) to reduce dimensionality.
- Reconstructed matrix to predict ratings for all books.
- Recommended top N books with highest predicted ratings.

Result:

- Generated high-coverage personalized recommendations for every user.
- Performed best in sparse environments due to its ability to generalize via latent dimensions.
- Best performance among all three techniques.

Example Output:

```
User A H Kobayashi => SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']
User A K P => SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']
User A Reviewer => SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']
User A Slater => SVD Recommendations: ['B000WSFBO0', 'B000R93D4Y', 'B001DOHZ5A', 'B002F3PPVE', 'B002BDT64A']
User A0089401235VSN3Z6F3HK => SVD Recommendations: ['B002RKRMSY', 'B000SN6IOQ', 'B000JQUT8S', 'B002RKSZJO', 'B000JMKXYW']
```

5. Evaluation and Comparisons:

```
Recommendations for User A H Kobayashi
Books already rated: ['1250787653']
User-User CF Recommendations: []
Item-Item CF Recommendations: []
SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']


Recommendations for User A K P
Books already rated: ['1450805752']
User-User CF Recommendations: []
Item-Item CF Recommendations: ['553508784', '110199715X', '545931908', '1250787653', '1635575583']
SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']


Recommendations for User A Reviewer
Books already rated: ['110199715X']
User-User CF Recommendations: []
Item-Item CF Recommendations: ['553508784', '545931908', '1250787653', '1450805752', '1635575583']
SVD Recommendations: ['547199457', '078525224X', '694003611', '545261244', '1635575605']
```

User: A H Kobayashi

Rated: ['1250787653']

- User-User CF: []

  No recommendations found due to lack of overlapping ratings.

- Item-Item CF: []

  Item similarity did not yield any results for this book.

- SVD Recommendations:

  ['547199457', '078525224X', '694003611', '545261244', '1635575605']

  SVD handled sparsity well and recommended diverse books.

User: A K P

Rated: ['1450805752']

- User-User CF: []

  Sparse profile, insufficient neighbors found.

- Item-Item CF:

  ['553508784', '110199715X', '545931908', '1250787653', '1635575583']

  Recommended based on similarity to the rated item.

- SVD Recommendations:

  ['547199457', '078525224X', '694003611', '545261244', '1635575605']

  Personalized and complete suggestions.

User: A Reviewer

Rated: ['110199715X']

- User-User CF: []
  No similar users found for collaborative filtering.
- Item-Item CF:
  ['553508784', '545931908', '1250787653', '1450805752', '1635575583']
  Effective suggestions based on the one rated book.
- SVD Recommendations:
  ['547199457', '078525224X', '694003611', '545261244', '1635575605']
  Successfully filled in likely preferences.

Insight:

SVD consistently produces recommendations even for sparse users, outperforming both User-User and Item-Item CF in coverage.

<u>Observations</u>

- User-User CF struggled due to lack of overlapping ratings.
- Item-Item CF performed reliably when at least one rating was present.
- SVD provided the best overall recommendations — consistent and broad.